MYP MATHEMATICS · YEAR 1

Mathematics for the MYP

Number systems and place value · Chapter 1 revision proof

1

SECOND EDITION · STUDENT COURSEBOOK · MYP 1 ONLY

01

Number systems and place value

Ten digits are enough to write any number we can name. Their power comes from position: the same symbol can mean seven, seventy or seven thousandths.

ENGAGE

A museum label says “about 49,000 objects”, while an inventory says “48,672 objects”. Which number would you put on a poster? Which would you use to check the collection? What changes when we round?

Framing the inquiry

Key concept: Form
Related concepts: Quantity, Representation
Global context: Orientation in space and time

Statement of inquiry: The position of a digit in a base-ten system encodes its value, so a small set of symbols can represent any magnitude when place and zero are used consistently.

Questions to guide us

- **Factual:** What is the value of a digit in a given position?
- **Conceptual:** Why is zero essential, and how can one number have several equivalent representations?
- **Debatable:** When is “about 2,000” more useful than the exact count?

ATL · Critical thinking. In order for a student to evaluate and improve a mathematical claim, students must test the claim on several cases, find a counterexample if one exists, and revise the statement so it is accurate.

GETTING READY

Learning intentions and prerequisite check

You are learning to

- Understand how natural numbers, whole numbers, integers and rational numbers are related.
- Know that the value of a digit depends on its place in a base-ten number.
- Write whole numbers and decimals in expanded form and return to standard form.
- Compare and order integers and decimals using place value and a number line.
- Round to a stated place and choose a reasonable level of accuracy in context.
- Estimate a result and explain whether the estimate is likely to be too high or too low.

Success criteria

I can classify a number and explain the role of zero.

I can state the value of any digit in a whole number or decimal.

I can move between standard form, expanded form and a number line.

I can order positive and negative numbers and justify the order.

I can round to a stated place and avoid an over-precise answer.

I can estimate, check exactly, and explain the difference.

READY CHECK

Try these without a calculator. They check the ideas this chapter builds on.

R1 Write “four thousand and sixty” using digits. READY

R2 Which is greater, 0.5 or 0.45? Explain using hundredths. READY

R3 Calculate 8×7. READY

R4 What is 100 more than 2,950? READY

R5 Write 7/10 as a decimal. READY

R6 Put −2, 0, 3 and −5 in order from least to greatest. READY

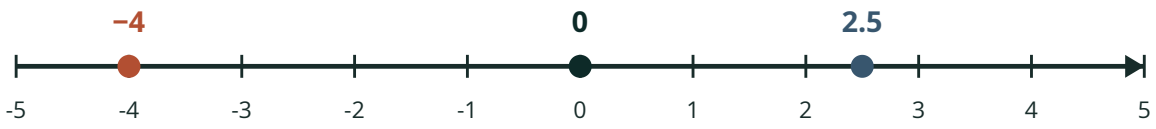
> **HOW TO USE THIS CHAPTER**
>
> Work through the examples before the practice. Use the diagrams to explain your thinking. Answers and student-facing reasoning are at the back of this proof. If more than two ready-check items are incorrect, review place value to hundreds and simple fractions before continuing.

1.1 · NUMBER SYSTEMS AND THE NUMBER LINE

Counting numbers, zero and negatives

Natural numbers are the counting numbers $1, 2, 3, \ldots$. In this book, **whole numbers** are the natural numbers together with zero: $0, 1, 2, 3, \ldots$. **Integers** include the whole numbers and their negatives: $\ldots, -3, -2, -1, 0, 1, 2, 3, \ldots$.

A number line gives every number a position. Values increase from left to right. Zero is the boundary between negative and positive numbers; it is neither positive nor negative.



Values increase from left to right. The point 2.5 has a precise position even though it is not an integer.

WORKED EXAMPLE 1

Classify -4, 0 and 12.

1. -4 is an integer and a rational number, but not whole or natural.
2. 0 is whole, an integer and rational, but not natural in this book.
3. 12 is natural, whole, an integer and rational.

Each number can belong to more than one family.

PRACTICE 1.1 · NUMBER FAMILIES AND ORDER

Show enough working or explanation that someone else can check your thinking.

1.1.1 State every family each number belongs to: natural, whole, integer, rational. (a) 12 (b) 0 (c) −4 (d) 5/8 (e) 1.2 FLUENCY

1.1.2 Write all the integers from −3 to 2 inclusive. FLUENCY

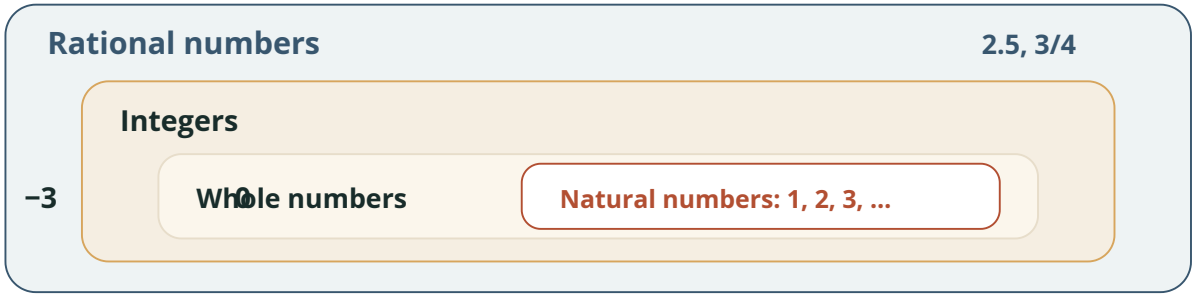
1.1.3 Replace each box with <, > or =. (a) −7 ☐ −2 (b) 0 ☐ −1 (c) −5 ☐ 5 FLUENCY

1.1.4 Order these temperatures from coldest to warmest: 3 °C, −2 °C, −7 °C, 0 °C, 5 °C. FLUENCY

1.1 · NUMBER SYSTEMS AND THE NUMBER LINE

Rational numbers and number families

A **rational number** can be written as a fraction $\frac{a}{b}$ where a and b are integers and $b \neq 0$. Terminating decimals such as 1.2 and fractions such as $\frac{5}{8}$ are rational. Every integer is rational because, for example, $-4 = \frac{-4}{1}$.



Natural numbers sit inside whole numbers; whole numbers sit inside integers; integers sit inside rational numbers, which also include fractions and terminating decimals.

MISCONCEPTION CHECK

"Zero is not a number because it means nothing." Zero is a number with a precise position. It also holds a place: 305 is not the same as 35.

WORKED EXAMPLE 2

Order $-7, -2, 0, 3, 5$ from least to greatest.

1. On a number line, more negative values lie further left.
2. Read from left to right: $-7, -2, 0, 3, 5$.

Coldest to warmest: $-7\,^\circ\text{C}, -2\,^\circ\text{C}, 0\,^\circ\text{C}, 3\,^\circ\text{C}, 5\,^\circ\text{C}$.

PRACTICE 1.1 · NUMBER FAMILIES AND ORDER

Show enough working or explanation that someone else can check your thinking.

1.1.5 Maya says, "Zero is a natural number because we can count zero items." Using this book's definition, explain whether her statement is correct. REASONING

1.1.6 Give a counterexample to the claim: "Every fraction is less than 1." REASONING

1.1.7 A lift starts at floor 1, travels to floor −2, then to floor 3. Describe each movement and the final position relative to the start. APPLICATION

1.1.8 Draw a number line from −5 to 5. Mark −4, 0 and 2.5. Explain why 2.5 is not an integer but still has a precise place. CHALLENGE

1.2 · PLACE VALUE AND THE ROLE OF ZERO

Position gives a digit its value

In a base-ten system, each position is worth ten times the position to its right. The decimal point separates whole-number places from fractional places. A zero can be a **placeholder**: it keeps the other digits in the correct positions.

Thousands	Hundreds	Tens	Ones	Tenths	Hundredths	Thousandths
3	0	4	8	3	0	7

decimal point

The highlighted 4 is in the tens place, so it is worth 40. The zeros in the hundreds and hundredths places are placeholders.

WORKED EXAMPLE 3

State the value of the 4 in $3{,}048.307$ and calculate 36.2×10.

1. The 4 is in the tens place, so its value is 40.
2. Multiplying by 10 shifts every digit one place left: $36.2 \times 10 = 362$.

The decimal point does not "move" by itself; the digits shift relative to it.

PRACTICE 1.2 · PLACE VALUE AND ZERO

Name the place before you state the value.

1.2.1 State the value of the underlined digit. (a) $6{,}7\underline{4}2$ (b) $23.5\underline{8}$ (c) $\underline{8}{,}106$ (d) $0.2\underline{0}4$ FLUENCY

1.2.2 Write the number represented by $4{,}000 + 600 + 30 + 2$. FLUENCY

1.2.3 Calculate. (a) 36.2×10 (b) 5.48×100 (c) $920 \div 10$ (d) $7.3 \div 100$ FLUENCY

1.2.4 In 2.06, what does the zero do? What number results if the zero is removed, and how much larger is it? REASONING

1.2 · PLACE VALUE AND THE ROLE OF ZERO

Zero, decimals and digit cards

Adding a zero to the right of a whole number multiplies it by 10, but adding a zero after a decimal may leave the value unchanged: $0.4 = 0.40$. Multiplying or dividing by powers of ten shifts digits, not just the decimal point.



Use each card once. A four-digit number cannot begin with 0.

Use each card once. A four-digit number cannot begin with 0.

WORKED EXAMPLE 4

Use 5, 0, 8 and 2 once each to make the greatest and least four-digit numbers.

1. Greatest: put the largest digits in the highest places: 8,520.
2. Least: the first digit cannot be 0, so start with 2, then use the smallest remaining digits: 2,058.

Greatest 8,520**; least** 2,058**.**

PRACTICE 1.2 · PLACE VALUE AND ZERO

Name the place before you state the value.

1.2.5 Which is larger, 3.5 or 3.05? Explain using place value, not the number of digits. REASONING

1.2.6 A warehouse receives 24 boxes with 100 pencils in each box. Write the total and state the value of the digit 2 in your answer. APPLICATION

1.2.7 Test the claim “adding a zero to the right of a number multiplies it by 10” on 4, 0.4 and 4.50. Then write a corrected rule. REASONING

1.2.8 Using 5, 0, 8 and 2 exactly once each, make the greatest and least four-digit numbers. Explain why the least cannot begin with 0. CHALLENGE

EXPLORATION · CRITERION B

How many numbers can the digits write?

How many positive whole numbers have exactly n digits? A number cannot begin with 0. Use a table, look for a pattern, then test your rule on a new case.

DIGITS, N	SMALLEST	LARGEST	HOW MANY?
1	1	9	9
2	10	99	
3	100	999	
4	1,000	9,999	
5			

EXPLORATION · HOW MANY NUMBERS CAN THE DIGITS WRITE?

Criterion B practice. Use a table, describe a rule, then verify it on a new case.

B1 Using exactly n digits, with repetition allowed and no leading zero, complete the count for n = 1, 2, 3 and 4. INVESTIGATION

B2 Describe how the total changes when one more digit is allowed. INVESTIGATION

B3 State a rule for n digits and test it for n = 5. INVESTIGATION

B4 Explain why the first digit has 9 choices but each later digit has 10. INVESTIGATION

B5 Challenge: repeat the investigation in base 5 using digits 0–4. Record three cases and state a rule. CHALLENGE

ATL · Critical thinking. In order for a student to verify a general rule, students must test it on a new case and explain why the rule works, not only show that it worked before.

1.3 · EXPANDED FORM AND EQUIVALENT REPRESENTATIONS

One number, several useful forms

Expanded form writes a number as the sum of the values of its digits. It makes the contribution of each place visible. Standard form is the usual compact way of writing the number.

WORKED EXAMPLE 5

Write 30.78 in expanded form, then return to standard form.

1. Identify each place: 3 tens, 0 ones, 7 tenths, 8 hundredths.
2. Expanded form: $30 + 0.7 + 0.08$.
3. Adding the parts gives 30.78 again.

$$30.78 = 30 + 0.7 + 0.08 = 3 \times 10 + 7 \times 0.1 + 8 \times 0.01.$$

WHY ZERO STILL MATTERS

$407.3 = 400 + 7 + 0.3$. The zero tens do not add value, but they keep the 4 in the hundreds place and the 7 in the ones place.

PRACTICE 1.3 · EXPANDED FORM

Move between standard form and expanded form without changing the value.

1.3.1 Write in expanded form. (a) 2,506 (b) 30.78 (c) 0.409 FLUENCY

1.3.2 Write as a standard number. (a) $4 \times 100 + 9 \times 10 + 6 \times 1$ (b) $6 \times 1 + 5 \times 0.1 + 8 \times 0.01$ (c) $2 \times 1{,}000 + 7 \times 10 + 3 \times 0.001$ FLUENCY

1.3.3 Fill the missing value. (a) 8,000 + ☐ + 40 + 5 = 8,745 (b) 0.6 + 0.07 + ☐ = 0.679 FLUENCY

1.3.4 Explain why $2 \times 0.1 + 5 \times 0.01$ is ten times $2 \times 0.01 + 5 \times 0.001$. REASONING

1.3 · EXPANDED FORM AND EQUIVALENT REPRESENTATIONS

Explain the structure, not just the answer

Equivalent representations have the same value but can make different features visible. A place-value chart shows positions; expanded form shows contributions; a number line shows order and distance.

WORKED EXAMPLE 6

Explain why $2 \times 0.1 + 5 \times 0.01$ is ten times $2 \times 0.01 + 5 \times 0.001$.

1. First expression: $0.2 + 0.05 = 0.25$.
2. Second expression: $0.02 + 0.005 = 0.025$.
3. Each digit has moved one place to the left, so each contribution is ten times as large.

$0.25 = 10 \times 0.025$.

PRACTICE 1.3 · EXPANDED FORM

Move between standard form and expanded form without changing the value.

1.3.5 Student A writes 407.3 as 400 + 7 + 0.3. Student B writes 4 × 100 + 0 × 10 + 7 × 1 + 3 × 0.1. Whose form shows the tens place more clearly? Do they have the same value? REASONING

1.3.6 A label gives a length as 5 × 10 + 3 × 1 + 4 × 0.1 centimetres. Write the length in standard form and name the unit of each term. APPLICATION

1.3.7 Create a four-digit number with 6 in the hundreds place and 0 in the tens place. Swap those two digits and explain how the value changes. CHALLENGE

1.3.8 Why can 1,000 be written with one nonzero term in expanded form while its place-value chart still has four positions? REASONING

COMMUNICATE

When you explain a representation, name the place, state its value, and say what the representation makes easy to see.

1.4 · COMPARING, ORDERING AND ROUNDING

Compare place by place

To compare decimals, align the decimal points and compare digits from the highest place downwards. Add trailing zeros if helpful: $0.5 = 0.50$. For negative numbers, the number further left on the number line is smaller.



Many decimals lie between 3.5 and 3.6. The zoomed line shows 3.501 and 3.512.

WORKED EXAMPLE 7

Order $0.7, 0.07, 0.77, 0.707$ from least to greatest.

1. Write with three decimal places: $0.700, 0.070, 0.770, 0.707$.
2. Compare: $0.070 < 0.700 < 0.707 < 0.770$.

$0.07, 0.7, 0.707, 0.77$.

PRACTICE 1.4 · COMPARE, ORDER AND ROUND

For rounding, state the target place and use the digit immediately to its right.

1.4.1 Insert <, > or =. (a) 0.5 ☐ 0.50 (b) 2.7 ☐ 2.68 (c) −3 ☐ −5 FLUENCY

1.4.2 Order from least to greatest: 0.7, 0.07, 0.77, 0.707. FLUENCY

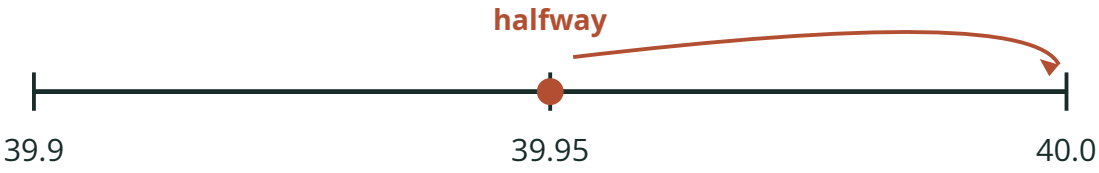
MISCONCEPTION CHECK

“More digits means a larger decimal.” Not always: $6.09 < 6.1$ because $6.1 = 6.10$.

1.4 · COMPARING, ORDERING AND ROUNDING

Round to a stated place

To round, identify the target place and look at the digit immediately to its right. If that digit is 5 or more, round up; otherwise round down. This book uses the usual convention that a 5 in the deciding place rounds up.



Rounding 39.95 to the nearest tenth gives 40.0 because the hundredths digit is 5 and the carry reaches the whole-number part.

WORKED EXAMPLE 8

Round 59.96 to the nearest tenth.

1. The target is the tenths place: 9. The deciding digit is 6.
2. Round 9 tenths up to 10 tenths, which is one whole.
3. The result is 60.0. The trailing zero records tenths.

$59.96 \approx 60.0$ **to the nearest tenth.**

PRACTICE 1.4 · COMPARE, ORDER AND ROUND

For rounding, state the target place and use the digit immediately to its right.

1.4.3 Round 8,647 to the nearest (a) hundred (b) ten. FLUENCY

1.4.4 Round 3.746 to the nearest hundredth and 0.897 to the nearest tenth. FLUENCY

1.4.5 Round 59.96 to the nearest tenth. Explain why the whole-number part changes. REASONING

1.4 · COMPARING, ORDERING AND ROUNDING

Rounding in context

A rounded number is not the exact value; it is a statement about a range. Always state the place or unit used. In accountability contexts, give the exact figure as well as any rounded headline.

WORKED EXAMPLE 9

Give three decimals that round to 6.5 to the nearest tenth, then describe all possible values.

1. Examples: $6.45, 6.50, 6.549$.
2. The lower boundary is 6.45 because the deciding digit is 5.
3. The upper boundary is just below 6.55, because 6.55 would round to 6.6.

All values from 6.45 inclusive up to but not including 6.55.

PRACTICE 1.4 · COMPARE, ORDER AND ROUND

For rounding, state the target place and use the digit immediately to its right.

1.4.6 One sign says a trail is 4.6 km; another says 4.56 km. Explain how both could be correct. REASONING

1.4.7 Give three decimals that round to 6.5 to the nearest tenth. Describe all possible values under the usual "5 rounds up" convention. CHALLENGE

1.4.8 A class raised 1,948 CFA francs and reports "about 2,000 CFA francs". Is this reasonable? What should an accountability report include? APPLICATION

1.5 · ESTIMATION AND SENSIBLE ACCURACY

Estimate first, then check

An **estimate** uses friendly numbers to predict a result or check whether an exact answer is reasonable. State the rounding you used. When one number is rounded up and another down, the estimate may be high or low; an exact check tells you which.

WORKED EXAMPLE 10

Estimate 58×31, then calculate exactly.

1. Round to the nearest ten: $60 \times 30 = 1{,}800$.
2. Exactly: $58 \times 31 = 1{,}798$.
3. The estimate is 2 high.

Estimate 1,800; exact 1,798.

PRACTICE 1.5 · ESTIMATE AND JUDGE ACCURACY

Estimate first, then check. Say what your estimate is for and whether it is above or below the exact result when you can tell.

1.5.1 Estimate 692 + 3,141 by rounding each number to the nearest hundred. Then calculate exactly and compare. FLUENCY

1.5.2 Estimate 58 × 31 by rounding each factor to the nearest ten. Then calculate exactly and compare. FLUENCY

1.5.3 Estimate 9.8 × 4.2 using 10 × 4. Calculate the exact product. Is the estimate above or below? Can you tell from the rounding alone? REASONING

1.5.4 Estimate 604 ÷ 29 using 600 ÷ 30. Then compare with the exact quotient to one decimal place. FLUENCY

REASONING HABIT

Ask: “What is this estimate for?” A quick headline, a budget check and an inventory may need different accuracy.

1.5 · ESTIMATION AND SENSIBLE ACCURACY

Choose accuracy for a purpose

Extra digits are not automatically more truthful. A calculation can produce many decimal places even when the original measurement was only accurate to the nearest centimetre. Report a level of accuracy that the context can support.

WORKED EXAMPLE 11

A library has 1,247,936 books. Suggest a figure for a short public notice.

1. A headline does not need the exact count.
2. “About 1,250,000 books” to the nearest ten thousand is readable.
3. An inventory or audit should use 1,247,936 exactly.

Match the accuracy to the purpose, and state the rounding used.

PRACTICE 1.5 · ESTIMATE AND JUDGE ACCURACY

Estimate first, then check. Say what your estimate is for and whether it is above or below the exact result when you can tell.

1.5.5 A shop records 397 visitors on Friday, 412 on Saturday and 289 on Sunday. Estimate the total to the nearest hundred, then calculate it exactly. APPLICATION

1.5.6 A runner completes 9.95 km and 10.08 km on two days. A poster says “about 20 km”. Is this reasonable? What accuracy is being used? REASONING

1.5.7 A library has 1,247,936 books. Suggest a rounded figure for a short public notice and state its accuracy. When should the exact figure be used? CHALLENGE

1.5.8 A calculation gives a length as 2.673913043 m. The measuring tape reads to the nearest centimetre. Suggest a sensible report and explain why the extra digits should not be presented as measured accuracy. CHALLENGE

CHECKPOINT

Before the application

Answer quickly, then explain one answer to a partner. Use the answers at the back to check both the result and the reasoning.

CHECKPOINT · BEFORE THE APPLICATION

Answer quickly, then explain one answer to a partner.

C1	What is the value of the 7 in 0.0705?	CHECK
C2	Calculate 4.06 × 100.	CHECK
C3	Round 39.95 to the nearest tenth.	CHECK
C4	Estimate 49 × 21 using friendly numbers.	CHECK

Common slips

- Forgetting a placeholder zero: 305 is not 35.
- Comparing decimals by counting digits.
- Rounding 39.95 to 39.9 instead of 40.0.
- Assuming an estimate must be high or low without checking.

Explain how you know

- Name the place before stating a digit's value.
- Use a number line to justify an order.
- State the target place before rounding.
- Compare an estimate with an exact result when possible.

APPLICATION · CRITERION D

A school noticeboard

Simulated context. A school in Cotonou is preparing a public noticeboard. The draft figures are below. They are classroom data, not official statistics.

FIGURE	DRAFT VALUE	POSSIBLE PURPOSE
Enrolment	1,284	Headline or planning
Library books	3,680	Inventory
Year opened	1998	History panel
Annual visits	10,473	Headline
Scholarship places	12	Application notice

APPLICATION · A SCHOOL NOTICEBOARD

Criterion D practice. This is a simulated dataset for a school in Cotonou. Choose forms that fit each purpose, justify accuracy, and check that the message makes sense.

D1 The draft figures are: enrolment 1,284; library books 3,680; year opened 1998; annual visits 10,473; scholarship places 12. Recommend a form for each and justify your choice. APPLICATION

D2 Another school has enrolment 1,327. If both enrolments are rounded to the nearest hundred, what does a reader see? Is the comparison fair? Suggest a better accuracy. APPLICATION

D3 What must the school clarify before publishing “10,473 visits”? APPLICATION

D4 Identify one figure that should not be rounded and one that is most sensitive to rounding. Explain both choices. APPLICATION

ATL · Communication. In order for a student to communicate mathematical choices for a real purpose, students must select an appropriate form and accuracy, justify the choice, and check that the message is clear to the intended audience.

ASSESSMENT PRACTICE · CRITERIA A AND C

A museum label

Simulated fact sheet. A museum collection has 48,672 objects, with 6,509 on display. The museum opened in 1894 and recorded 214,830 visitors last year.

CRITERION A · KNOWING AND UNDERSTANDING
Use accurate place value, expanded form and rounding.

CRITERION C · COMMUNICATING
Write for visitors aged 10–14 and adults who want a quick overview. Use mathematical language, forms and a clear structure.

ASSESSMENT PRACTICE · A MUSEUM LABEL

Criteria A and C. Use accurate mathematics and clear communication. Write for visitors aged 10–14 and adults who want a quick overview.

S1 The fact sheet gives 48,672 objects in the collection and 6,509 on display. State the value of the 8 in 48,672 and the 6 in 6,509. ASSESSMENT

S2 Write 6,509 in expanded form. ASSESSMENT

S3 Round 48,672 to the nearest thousand and 214,830 annual visitors to the nearest ten thousand. ASSESSMENT

S4 Write a label of no more than 70 words using the collection size, display count, opening year 1894 and annual visitors. Include one rounded value and one comparison. ASSESSMENT

S5 Add a two-sentence note explaining one rounding choice and one risk of rounding too heavily. ASSESSMENT

BEFORE YOU FINISH

Check that every rounded value states its accuracy, that the opening year is exact, and that your comparison is supported by the numbers.

CHAPTER REVIEW

What you should now be able to do

Key ideas

- Number families are nested: natural ⊂ whole ⊂ integer ⊂ rational.
- Place value gives each digit a value; zero can hold a place.
- Expanded form shows each digit's contribution.
- Compare decimals place by place, not by counting digits.
- Rounding records a range and a chosen accuracy.
- Estimates are useful when checked and matched to purpose.

Vocabulary

natural number · whole number · integer · rational number · place value · placeholder · decimal point · expanded form · standard form · number line · rounding · estimate · accuracy

MIXED PRACTICE 1

Do not use a calculator unless a question asks for one. Show a method or explanation where requested.

M1 Write the value of the digit 8 in (a) 3,812 (b) 20.83 (c) 0.081. FLUENCY

M2 Write 5,040.207 in expanded form. FLUENCY

M3 Write $7 \times 1{,}000 + 3 \times 10 + 6 \times 0.01$ as a decimal. FLUENCY

M4 Round 64,782 to the nearest thousand, hundred and ten. FLUENCY

M5 Round 0.496 to the nearest hundredth and 12.95 to the nearest tenth. FLUENCY

M6 Calculate (a) 0.64×100 (b) $8.2 \div 10$ (c) $3.5 \times 1{,}000$. FLUENCY

MIXED PRACTICE 1

Fluency and reasoning

MIXED PRACTICE 1

Do not use a calculator unless a question asks for one. Show a method or explanation where requested.

M7 Order from least to greatest: −2.5, 0.25, −0.5, 2.05. FLUENCY

M8 Insert <, > or =. (a) 0.08 ☐ 0.8 (b) 1.30 ☐ 1.3 (c) −6 ☐ −2 FLUENCY

M9 A whole number rounds to 4,700 to the nearest hundred. Give three possible values and describe the full range. REASONING

M10 A student says 6.09 is larger than 6.1 because it has more digits. Correct the comparison. REASONING

M11 A temperature changes from −3 °C to 4 °C. Which is colder, and by how many degrees do they differ? REASONING

M12 A stock list gives 12,480 items. The manager says “about 12,000”. Is this an underestimate or overestimate, and by how much? Suggest a more informative rounded statement. APPLICATION

EXPLAIN HOW YOU KNOW

Choose one answer and write a short justification using place value, a number line or an exact check.

MIXED PRACTICE 1 · REFLECTION

Challenge and reflect

MIXED PRACTICE 1

Do not use a calculator unless a question asks for one. Show a method or explanation where requested.

M13 Explain the different value of the 5 in 4,507 and 457. REASONING

M14 Estimate 3,912 + 4,087 + 2,998 to the nearest thousand. Is the estimate above or below the exact total? Check. REASONING

M15 A number is $6 \times 100 + 4 \times 10 + \square \times 1$. It is even and lies between 646 and 650. Find the missing digit and explain why it is unique. CHALLENGE

M16 Why do 2.60 and 2.6 have the same value, but 2.60 may be a more appropriate measurement report? REASONING

M17 Write a number between 3.5 and 3.6 with three decimal places. Then write a second number in the same interval closer to 3.5. Justify the order. CHALLENGE

M18 Find four-digit numbers whose digits sum to 12, whose thousands digit is 3, and whose tens digit is twice the ones digit. Find all possibilities. CHALLENGE

Reflection

I can explain the role of zero with an example and a counterexample.

I can move between standard form, expanded form and a number line.

I can round and state the range of values a rounded number could represent.

I can choose an accuracy for a real purpose and justify it.

One idea I would explain differently next time: _____

One question I still have: _____

CHAPTER 1 · ANSWERS AND REASONING

Answers · Ready check and Practice 1.1

READY CHECK

R1 **4,060**
The 4 is in the thousands place and the 6 is in the tens place; zeros hold the hundreds and ones places.

R2 **0.5**
0.5 = 0.50, and 50 hundredths is greater than 45 hundredths.

R3 **56**
Eight groups of seven make 56.

R4 **3,050**
Adding one hundred to 29 hundreds gives 30 hundreds, so the thousands digit increases.

R5 **0.7**
Seven tenths is written 0.7.

R6 **−5, −2, 0, 3**
On a number line, values increase from left to right.

PRACTICE 1.1 · NUMBER FAMILIES AND ORDER

1.1.1 **(a) natural, whole, integer, rational. (b) whole, integer, rational. (c) integer, rational. (d) rational. (e) rational.**
Natural numbers start at 1 in this book; whole numbers add 0; integers add negatives; rational numbers can be written as a fraction of integers with nonzero denominator.

1.1.2 **−3, −2, −1, 0, 1, 2**
Inclusive means both endpoints are included.

1.1.3 **(a) < (b) > (c) <**
The smaller number lies further left on the number line.

1.1.4 **−7 °C, −2 °C, 0 °C, 3 °C, 5 °C**
Coldest means least, so order the numbers from left to right on a number line.

1.1.5 **Not under this book's convention. Zero is whole, an integer and rational, but natural numbers start at 1.**
Some traditions include 0 in the natural numbers, so a definition must be stated. This book uses natural numbers = 1, 2, 3, ...

1.1.6 **For example, 5/4 = 1.25, which is greater than 1. Also 4/4 = 1.**
A fraction can be greater than or equal to 1 when the numerator is at least the denominator.

1.1.7 **Down 3 floors, then up 5 floors; final position is 2 floors above the start.**
1 to −2 is a change of −3; −2 to 3 is a change of +5; net change +2.

1.1.8 **2.5 lies halfway between 2 and 3. It is rational, not an integer.**
Integers are whole numbers and their negatives; halves between integers are rational numbers.

CHAPTER 1 · ANSWERS AND REASONING

Answers · Practice 1.2 and Exploration

PRACTICE 1.2 · PLACE VALUE AND ZERO

1.2.1 **(a) 40 (b) 0.08 (c) 8,000 (d) 0**
The places are tens, hundredths, thousands and hundredths respectively. A zero digit contributes zero but still holds its place.

1.2.2 **4,632**
Combine the thousands, hundreds, tens and ones.

1.2.3 **(a) 362 (b) 548 (c) 92 (d) 0.073**
Multiplying by 10 shifts digits one place left; multiplying by 100 shifts two places. Division shifts digits right.

1.2.4 **It holds the tenths place. Removing it gives 2.6, which is 0.54 larger.**
2.06 has 0 tenths and 6 hundredths; 2.6 has 6 tenths. 2.60 − 2.06 = 0.54.

1.2.5 **3.5**
3.5 = 3.50. Both have 3 ones, but 5 tenths is greater than 0 tenths.

1.2.6 **2,400 pencils; the 2 is worth 2,000.**
24 × 100 = 2,400. The 2 is in the thousands place.

1.2.7 **4 ☒ 40 multiplies by 10; 0.4 ☒ 0.40 and 4.50 ☒ 4.500 do not change the value. Correct rule: multiplying by 10 shifts every digit one place left. Appending a zero only guarantees this for whole numbers written without a decimal part.**
Trailing zeros after a decimal point add no value; they may record precision.

1.2.8 **Greatest 8,520; least 2,058.**
Put the largest digits in the highest places for the greatest number. For the least four-digit number, the thousands digit must be the smallest nonzero digit, 2.

EXPLORATION · HOW MANY NUMBERS CAN THE DIGITS WRITE?

B1 **9; 90; 900; 9,000.**
The first digit has 9 choices; each later digit has 10 choices.

B2 **It is multiplied by 10.**
Each existing number can be extended by any of 10 final digits.

B3 **9 × 10^(n−1); for n = 5 the count is 90,000.**
There are 9 choices for the first digit and 10 choices for each of the remaining n−1 digits.

B4 **The first digit cannot be 0, so it can be 1–9. Later digits can be 0–9.**
A leading zero would not create an n-digit positive whole number.

B5 **For n = 1, 2, 3 the counts are 4, 20, 100. Rule: 4 × 5^(n−1).**
The first digit has 4 nonzero choices; each later digit has 5 choices.

CHAPTER 1 · ANSWERS AND REASONING

Answers · Practices 1.3 and 1.4

PRACTICE 1.3 · EXPANDED FORM

1.3.1 **(a) 2,000 + 500 + 6 (b) 30 + 0.7 + 0.08 (c) 0.4 + 0.009**
Each nonzero digit is multiplied by its place value; zero places may be omitted.

1.3.2 **(a) 496 (b) 6.58 (c) 2,070.003**
Keep zeros in any missing places so the other digits stay in the correct positions.

1.3.3 **(a) 700 (b) 0.009**
The missing parts are 7 hundreds and 9 thousandths.

1.3.4 **The first expression is 0.25 and the second is 0.025. Each digit in the first is one place to the left, so each contributes ten times as much.**
0.25 ÷ 0.025 = 10.

1.3.5 **Student B shows the zero tens explicitly. Both forms have the same value, 407.3.**
Adding a zero term does not change a sum, but it can reveal structure.

1.3.6 **53.4 cm. The terms are 50 cm, 3 cm and 0.4 cm.**
5 tens of centimetres + 3 centimetres + 4 tenths of a centimetre.

1.3.7 **Sample: 1,602 becomes 1,062, a decrease of 540. Other valid starting numbers are possible.**
Moving 6 from hundreds to tens loses 600 − 60 = 540.

1.3.8 **1,000 = 1 × 1,000. The hundreds, tens and ones are all zero, but those positions are still needed to show the 1 is in the thousands place.**
Zero terms contribute no value but hold places.

PRACTICE 1.4 · COMPARE, ORDER AND ROUND

1.4.1 **(a) = (b) > (c) >**
Compare place by place; for negatives, the number further right is greater.

1.4.2 **0.07, 0.7, 0.707, 0.77**
Write them as 0.070, 0.700, 0.707, 0.770 to compare thousandths.

1.4.3 **(a) 8,600 (b) 8,650**
For hundreds, the tens digit 4 rounds down; for tens, the ones digit 7 rounds up.

1.4.4 **3.75 and 0.9**
The thousandths digit 6 rounds the hundredths up; the hundredths digit 9 rounds the tenths up with a carry.

1.4.5 **60.0. The hundredths digit 6 rounds 9 tenths up to 10 tenths, which is one whole.**
Keep the .0 to show the answer is to the nearest tenth.

1.4.6 **4.56 km rounds to 4.6 km to the nearest tenth.**
The signs report the same length at different accuracies.

1.4.7 **Examples: 6.45, 6.50, 6.549. All values from 6.45 inclusive up to but not including 6.55 round to 6.5.**
6.45 is the lower midpoint; 6.55 would round to 6.6.

1.4.8 **Yes; 1,948 rounds to 2,000 to the nearest thousand. An accountability report should give the exact total, 1,948 CFA francs, and state any rounding used.**
The hundreds digit is 9, so 1,948 rounds up to 2,000 to the nearest thousand. “About 2,000” is reasonable for a quick headline, but an accountability report should keep the exact total and state the rounding.

CHAPTER 1 · ANSWERS AND REASONING

Answers · Practice 1.5 and Checkpoint

PRACTICE 1.5 · ESTIMATE AND JUDGE ACCURACY

1.5.1 **Estimate 700 + 3,100 = 3,800. Exact 3,833, so the estimate is 33 low.**
692 rounds up by 8; 3,141 rounds down by 41; the net change is −33.

1.5.2 **Estimate 60 × 30 = 1,800. Exact 1,798, so the estimate is 2 high.**
Rounding one factor up and one down does not guarantee the direction; the exact check shows it.

1.5.3 **Estimate 40; exact 41.16, so the estimate is below. Mixed rounding does not by itself guarantee the direction.**
Round 9.8 up to 10 and 4.2 down to 4: 10 × 4 = 40. Exactly, 9.8 × 4.2 = 41.16. One factor was rounded up and the other down, so the direction of the error is not guaranteed; the exact check shows this estimate is low.

1.5.4 **Estimate 20. Exact quotient is about 20.8, so the estimate is low.**
604 ÷ 29 = 20.793…; rounding the divisor down and dividend down has competing effects.

1.5.5 **Estimate 400 + 400 + 300 = 1,100. Exact 1,098; estimate is 2 high.**
The rounded values differ from the exact values by +3, −12 and +11.

1.5.6 **Yes. The exact total is 20.03 km, which rounds to 20 km to the nearest kilometre.**
The poster gives a useful whole-kilometre summary, not an exact distance.

1.5.7 **Sample: about 1,250,000 books, to the nearest ten thousand. Use the exact figure in an inventory or audit.**
The rounded figure is easier to read; the exact figure is needed when individual items must be accounted for.

1.5.8 **About 2.67 m is a sensible centimetre-level report. The extra digits come from calculation, not from measurement precision; a full accuracy claim would need the calculation and measurement uncertainties.**
One centimetre is 0.01 m. Do not imply the tape measured billionths of a metre.

CHECKPOINT · BEFORE THE APPLICATION

C1 **0.07**
The 7 is in the hundredths place.

C2 **406**
The digits shift two places left.

C3 **40.0**
The hundredths digit 5 rounds the tenths up, carrying into the ones.

C4 **About 50 × 20 = 1,000.**
The exact product is 1,029, so the estimate is close and slightly low.

CHAPTER 1 · ANSWERS AND REASONING

Answers · Application and Assessment practice

APPLICATION · A SCHOOL NOTICEBOARD

D1 **Sample: enrolment about 1,300 for a headline; library books 3,680 for inventory; year opened 1998 exactly; visits about 10,000 for a headline; scholarship places 12 exactly because the number is small and consequential.**
Accuracy should match purpose. Small counts and identifiers should not be rounded merely for style.

D2 **Both appear as about 1,300, hiding a difference of 43 students. To the nearest ten, 1,280 and 1,330 preserve the comparison better.**
1,327 − 1,284 = 43. Rounding to hundreds maps both into the same hundred.

D3 **Whether it counts visits or unique visitors, the period covered, and whether repeat visits by the same person are counted more than once.**
A number is interpretable only when the counted unit and collection rule are clear.

D4 **The year opened, 1998, should not be rounded because it identifies a date. The 12 scholarship places are sensitive: rounding to the nearest ten gives 10, losing 2 places, about 17% of the total.**
The same rounding rule can be harmless for a large headline and misleading for a small decision-making count.

ASSESSMENT PRACTICE · A MUSEUM LABEL

S1 **8,000 and 6,000.**
The 8 is in the thousands place; the 6 is in the thousands place.

S2 **6,000 + 500 + 9, or 6 × 1,000 + 5 × 100 + 0 × 10 + 9 × 1.**
The zero tens may be omitted or shown to reveal the place structure.

S3 **49,000 objects and 210,000 visitors.**
The hundreds digit 6 rounds the thousands up; the thousands digit 4 rounds the ten-thousands down.

S4 **Sample: “The collection began in 1894. Today it holds about 49,000 objects, with 6,509 on display—fewer than one in seven. Last year the museum welcomed about 210,000 visitors. Come and see how a small display can tell a large story.”**
The label uses the opening year exactly, rounds the collection and visitor figures for readability, and keeps the display count exact. The comparison is valid because 6,509 is less than one seventh of 48,672. It stays within 70 words.

S5 **Sample: “I rounded the collection to the nearest thousand so the headline is quick to read. Rounding the display count too heavily could hide how few objects visitors can actually see.”**
The note connects accuracy to audience and meaning.

CHAPTER 1 · ANSWERS AND REASONING

Answers · Mixed practice 1 (M1–M9)

MIXED PRACTICE 1

M1 **(a) 800 (b) 0.8 (c) 0.08**
The places are hundreds, tenths and hundredths.

M2 **5,000 + 40 + 0.2 + 0.007**
The hundreds, ones and hundredths digits are zero.

M3 **7,030.06**
Keep zeros in the hundreds, ones and tenths places.

M4 **65,000; 64,800; 64,780**
Inspect the digit immediately to the right of each target place.

M5 **0.50 and 13.0**
Both involve a carry. Keep the final zero to show the requested place.

M6 **(a) 64 (b) 0.82 (c) 3,500**
Shift digits left for multiplication by powers of ten and right for division.

M7 **−2.5, −0.5, 0.25, 2.05**
Negative values lie left of zero; among negatives, the larger magnitude is smaller.

M8 **(a) < (b) = (c) <**
Compare by place value and number-line position.

M9 **Examples: 4,650, 4,700, 4,749. Whole-number range: 4,650 to 4,749 inclusive.**
The midpoint below is 4,650; the midpoint above is 4,750, which would round to 4,800.

CHAPTER 1 · ANSWERS AND REASONING

Answers · Mixed practice 1 (M10–M18)

MIXED PRACTICE 1

M10 **6.09 < 6.1 because 6.1 = 6.10, and 9 hundredths is less than 10 hundredths.**
Align place values instead of counting digits.

M11 **−3 °C is colder; the difference is 7 °C.**
From −3 to 0 is 3 degrees, and from 0 to 4 is 4 degrees.

M12 **Underestimate by 480. “About 12,500” to the nearest hundred is more informative.**
12,480 − 12,000 = 480; 12,480 rounds to 12,500 to the nearest hundred.

M13 **In 4,507 the 5 is worth 500; in 457 it is worth 50. The zero in 4,507 keeps the 5 in the hundreds place.**
The same digit has different value in different positions.

M14 **Estimate 4,000 + 4,000 + 3,000 = 11,000. Exact total 10,997, so the estimate is 3 high.**
The rounded values differ by +88, −87 and +2, net +3.

M15 **8, giving 648.**
The only even whole number strictly between 646 and 650 is 648.

M16 **The trailing zero adds no value, but it records accuracy to the nearest hundredth rather than the nearest tenth.**
Numerical equality and reported precision are different ideas.

M17 **Sample: 3.512 and 3.501. Both lie between 3.500 and 3.600, and 3.501 is closer to 3.5.**
Compare thousandths: 3.500 < 3.501 < 3.512 < 3.600.

M18 **3,900; 3,621; 3,342; 3,063.**
Let the ones digit be u. Then tens is 2u and hundreds is 9 − 3u. The digit constraints allow u = 0, 1, 2, 3.